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Paleomagnetic results from the Early Cretaceous Zenong Group volcanic rocks, Cuoqin, Tibet, and their paleogeographic implications

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ABSTRACT

A paleomagnetic study has been carried out on the Early Cretaceous Zenong Group volcanic rocks (ca. 110–130 Ma) near Cuoqin town in the central Lhasa terrane. Stepwise thermal demagnetization up to 680 °C isolated a stable high-temperature component (HTC) including antipodal dual polarities. The HTC directions passed both a fold test at 99% confidence and a reversal test at 95% confidence, suggesting a primary origin. The tilt-corrected average direction for 18 sites is $D = 327.0°$, $I = 35.7°$, $\kappa = 59.3°$, $\alpha_{95} = 4.5°$, which corresponds to a paleopole at 58.2° N, 341.9° E ($A_{95} = 4.6°$), yielding a paleolatitude of $19.8° \pm 4.6°$ N for the study area. Our results, combined with previous paleomagnetic data, suggest that southern Tibet was at a mean paleolatitude of 20° N during the Cretaceous and Paleogene. When compared with the apparent polar wander paths of Asia, the difference suggests that only ca. 870 km north–south crustal shortening has accumulated between the Lhasa terrane and stable Asia since ca. 110 Ma. When comparing the paleolatitudes of the Lhasa terrane and India, the results indicate that the width of the Neotethys was 6720 ± 690 km during the Early Cretaceous.

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1. Introduction

The India–Asian collision and continued plate convergence in the Cenozoic Era lead to not only intracontinental deformation in Asia (Molnar and Tapponnier, 1975; Patriat and Achache, 1984; Hendrix et al., 1994; Matte et al., 1996; Meyer et al., 1998; Cao et al., 2011), but also uplift of the Tibetan Plateau (Xia et al., 2011). The latter may have had major repercussions on drastic changes of Asia climate (Molnar et al., 1993, 2010; Ramstein et al., 1997). The Lhasa terrane is located between the Himalayan terranes (Hodges, 2000) to the south and the Qiangtang terrane to the north (Fig. 1), and joined the growing Asian continent in the latest Jurassic–earliest Cretaceous (Chen et al., 1993; Matte et al., 1996; Metcalfe, 1996, 2006; Halim et al., 1998; Wu et al., 2002; Kapp et al., 2005, 2007; DeCelles et al., 2007). Because the Lhasa terrane was the southern margin of Asia prior to the Cenozoic India–Asian collision, Cretaceous paleomagnetic data from the Lhasa terrane are essential to reconstruct pre-collisional paleogeography. This, in turn, is key to understanding the India–Asian collision process and to estimating the magnitude of intracontinental deformation within Asia. Despite several paleomagnetic studies of Cretaceous rocks from the Lhasa terrane (Pozzi et al., 1982; Westphal et al., 1983; Achache et al., 1984; Lin and Watts, 1988; Chen et al., 1993; Sun et al., 2008; Tan et al., 2010), they have not converged on a consensus regarding the Cretaceous paleolatitude of the southern margin of Asia. Consequently, intra-Asia shortening estimates based on paleomagnetic data from more than 1500 km (Achache et al., 1984; Lin and Watts, 1988; Chen et al., 1993; Sun et al., 2008) to only a few hundred kilometers (Tan et al., 2010). Moreover, these studies suggest a north–south width of the Neotethyan Ocean ranging from 32 to 60° of latitude (Besse and Courtillot, 1988; Klootwijk et al., 1992; Van der Voo et al., 1999; Li et al., 2002). In addition, Cretaceous paleomagnetic investigations from the Lhasa terrane are not evenly distributed in space. All the studies except for the one reported by Chen et al. (1993) are confined to the eastern part of the Lhasa terrane (Fig. 1). Considering questions earlier, together with the fact that some Cretaceous paleomagnetic results from the Lhasa terrane (1) had no robust field tests and (2) might be affected by inclination shallowing due to compaction (Tan et al., 2010), we carried out a new paleomagnetic study for the Early Cretaceous volcanic rocks near Cuoqin area, in the central part of the Lhasa terrane.

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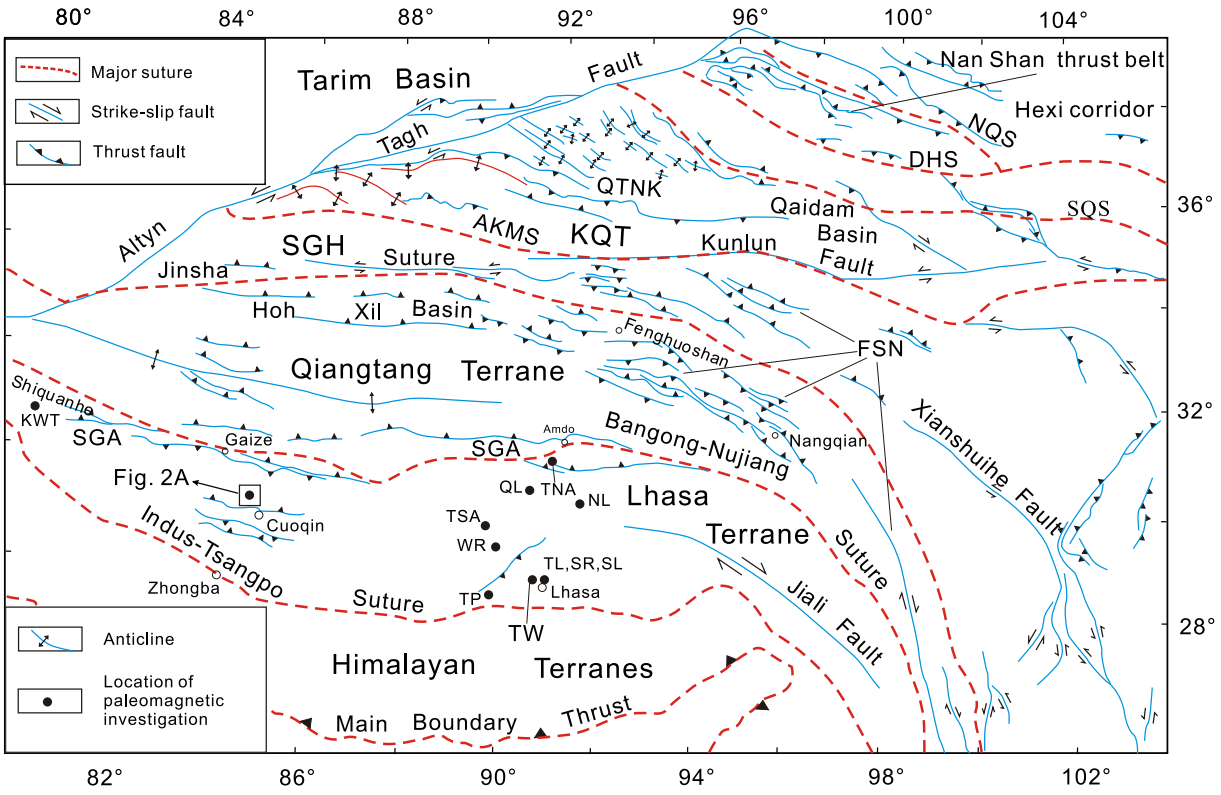


Fig. 1. (A) Tectonic sketch map of central Asia with the major active strike-slip faults modified after Yin and Harrison (2000). Closed circle shows the location of Cretaceous paleomagnetic investigations in the Lhasa terrane (abbreviation see Table 2). SGH Songpan–Ganzi–Hoh Xil Terrane; AKMS = Ayimaqin–Kunlun–Muztagh Suture; KQT = Kunlun–Qaidam Terrane; SQS = South–Qilian Suture; DHS = Danghenan Shan Suture; NQS = North–Qilian Suture; SGA = Shiquanhe–Gaize–Anduo thrust system; FSN = Fenghuo Shan–Nangqian fold and thrust belt; QTNK = Qimen Tagh–North Kunlun thrust system.

2. Geological setting and sampling

The Tibetan plateau can be divided primarily into five tectonic units that are, from south to north, the Himalayan, Lhasa, Qiangtang, Songpan–Ganze–Hoh Xil and Kunlun–Qaidam terranes (Fig. 1). These terranes are separated by sutures: the Indus–Tsangpo suture, which is mostly a Cretaceous to Tertiary suture (Tapponnier et al., 1986; Zhang et al., 2010), separates India from the Lhasa terrane. The Lhasa and Qiangtang terranes are separated by the Bangong–Nujiang suture, whereas the Jinsha suture separates the Qiangtang and Songpan–Ganze–Hoh Xil terranes. Finally, the Songpan–Ganze–Hoh Xil is separated from the Kunlun–Qaidam by the Ayimaqin–Kunlun–Muztagh suture (Fig. 1). In the central and northern area of the Lhasa terrane, the Early Cretaceous Zenong Group volcanic–sedimentary strata are widely exposed and have an average thickness more than 1000 m. Their lower part is composed of volcanic lavas mixed with pyroclastic rocks, and the upper part consists of volcaniclastic sedimentary rocks, normal volcanic conglomerate mixed lavas and pyroclastic rocks (Zhu et al., 2006). An extensive Cretaceous granite (104–106 Ma) intruded into the volcanic rocks of the Zenong Group (1:250,000 scale Cuoqin regional geological survey report (H45C001001), 2003). The zircon U–Pb ages reported by Zhu et al. (2008) indicated that the Zenong Group volcanism began at ca. 130 Ma and ceased at ca. 110 Ma.

We have collected 207 core samples (19 sites) from the Early Cretaceous Zenong Group volcanic rocks distributed at two localities separated by ca. 25 km (Fig. 2A). Thirty-two core samples (3 sites) were collected from 3 lavas located at ca. 20 km southwest of Daxiong town (31°18.8′ N, 84°49.6′ E). One hundred seventy-five samples (16 sites) were sampled from other localities near Daxiong town (31°22.2′ N–31°22.3′ N, 85°05′ E–85°06′ E) (Fig. 2A). Generally, each sampling site consists of at least 10 independently drilled samples that span several meters of stratigraphic thickness and cover at least one lava or multiple tuff layers (Fig. 2B, C, D). All samples were collected with a portable drill and oriented using both a magnetic and a solar compass.

3. Laboratory techniques and measurements

Standard 2.5-cm-diameter cores were cut into 2.2-cm-long specimens in the laboratory. Remanent magnetization measurements were carried out with 2 G-755R and 2 G-755-4 K cryogenic magnetometers, installed in the Institute of Geomechanics in Beijing (IG) and China University of Geosciences, Beijing (CUGB), respectively. Paleomagnetic specimens were thermally demagnetized up to 580 °C or 680 °C in ASC-TD 48 furnaces with an internal residual field less than 10 nT. 2 G-755R cryogenic magnetometer and the oven installed in IG are hosted inside large Helmholtz coils and samples were further protected from magnetic contamination with a shielded box during the experimental procedure. 2 G-755-4 K cryogenic magnetometer and the oven in CUGB are located in a shielded room with residual fields less than 300 nT. Demagnetization temperature intervals are generally large (50–30 °C) in the low temperature part, and become smaller (20–5 °C) at higher temperatures. In order to better understand magnetic mineralogy and ensure reliability of paleomagnetic data, acquisition of isothermal remanent magnetization (IRM), back-field demagnetization of saturation IRM (SIRM), and thermal demagnetization of the three-axis IRM (Lowrie, 1990) were performed on some representative specimens. Characteristic remanent magnetization (ChRM) directions of all the specimens were determined using principal component analysis (Kirschvink, 1980). Site-mean directions were calculated using Fisherian statistics (Fisher, 1953). Paleomagnetic data were analyzed using R. Enkin's and J.P. Cogné's computer program packages (Enkin, 1990; Cogné, 2003).

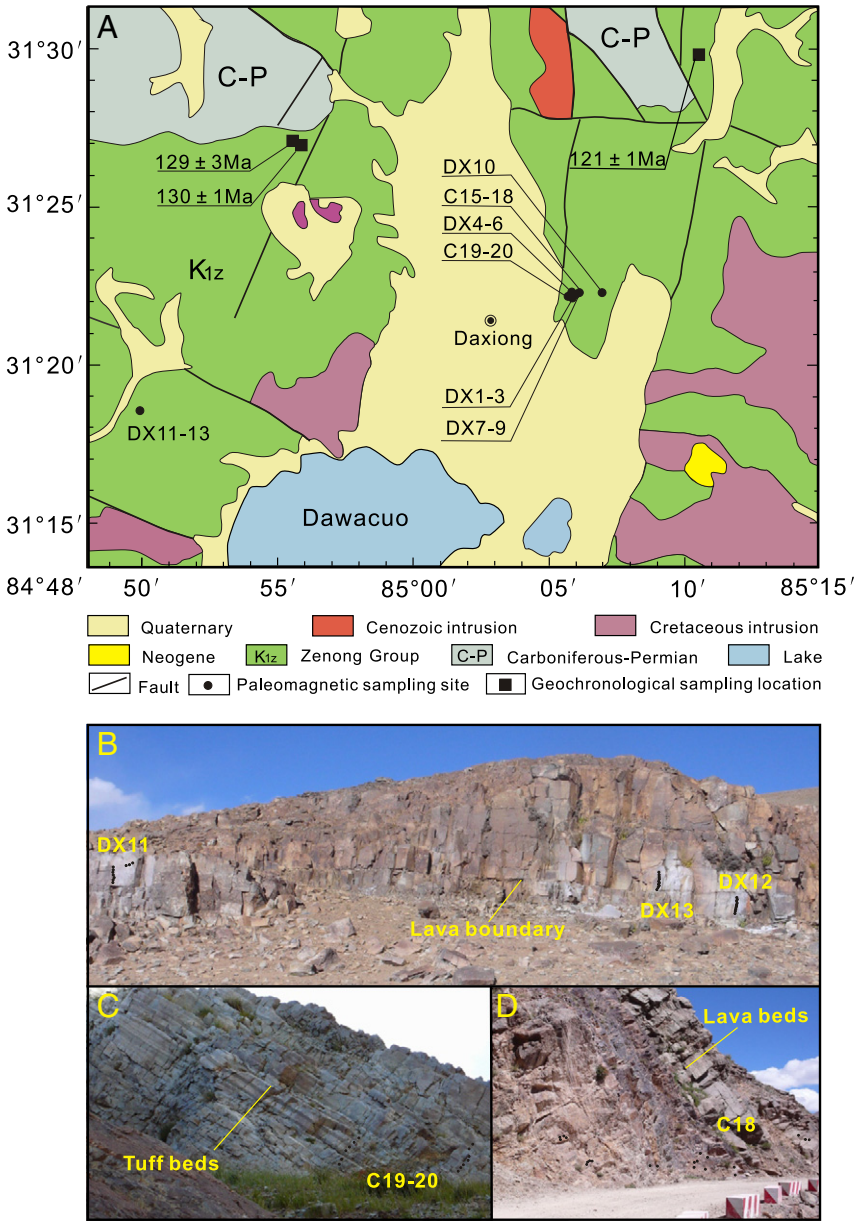


Fig. 2. (A) Simplified geological map of paleomagnetic sites location (closed circle). Closed rectangle indicates geochronological sites (Zhu et al., 2008). (B)–(D) Photographs showing the field outcrops in the studied section.

4. Paleomagnetic results

4.1. Tuff specimens

Sixty-three core samples covering 6 sites, which include sites C17, C19, C20, DX3, DX4, and DX5, were collected from tuff. A representative Zijderveld plot for tuff specimens is shown in Fig. 3A. After removing an unstable viscous magnetization component (VRM) by ca. 150 °C, a high-temperature magnetization component (HTC) that decayed toward the origin can be isolated between 200 °C and 580 °C. Its normalized intensity curve shows a sharp drop below 100 °C, and then a slow decay to less than 5% of the initial value after thermal demagnetization at 580 °C. The IRM acquisition curve of tuff specimens shows a continuous increase, and it is far from saturation even at 2.6 T. By applying the reverse field on the same specimen, there is an inflection point at the field of 65 mT and the maximum coercive force is 650 mT. These results suggest that high coercivity magnetic carriers are dominant, and coexist with low coercivity components (Fig. 4A). The Lowrie (1990) test shows that the hard (0.4–2.6 T) and medium (0.12–0.4 T) components unblocked at 100 °C are goethite, and the soft component (<0.12 T) presents clear evidence for magnetite unblocked at around 580 °C (Fig. 5A, B). The thermal demagnetization and rock-magnetic results mentioned earlier indicate that the unstable VRM is carried by goethite, and the HTC decayed to the origin is carried by magnetite.

4.2. Lava specimens

A total of 134 core samples from 13 sites were drilled from lava flows (C15, C16, C18, DX1, DX2, DX6–DX13). Most lava specimens have no a stable low-temperature component close to the present field, but between 350 °C and 660–680 °C all the lava specimens show a consistent HTC that decays toward the origin and yield a antipodal dual polarities (Fig. 3B, C). As shown in Fig. 4B, the IRM intensity of lava specimens present an abrupt increase below 1.3 T and then gradual increase until 2.6 T; saturation is not fully reached at 2.6 T.

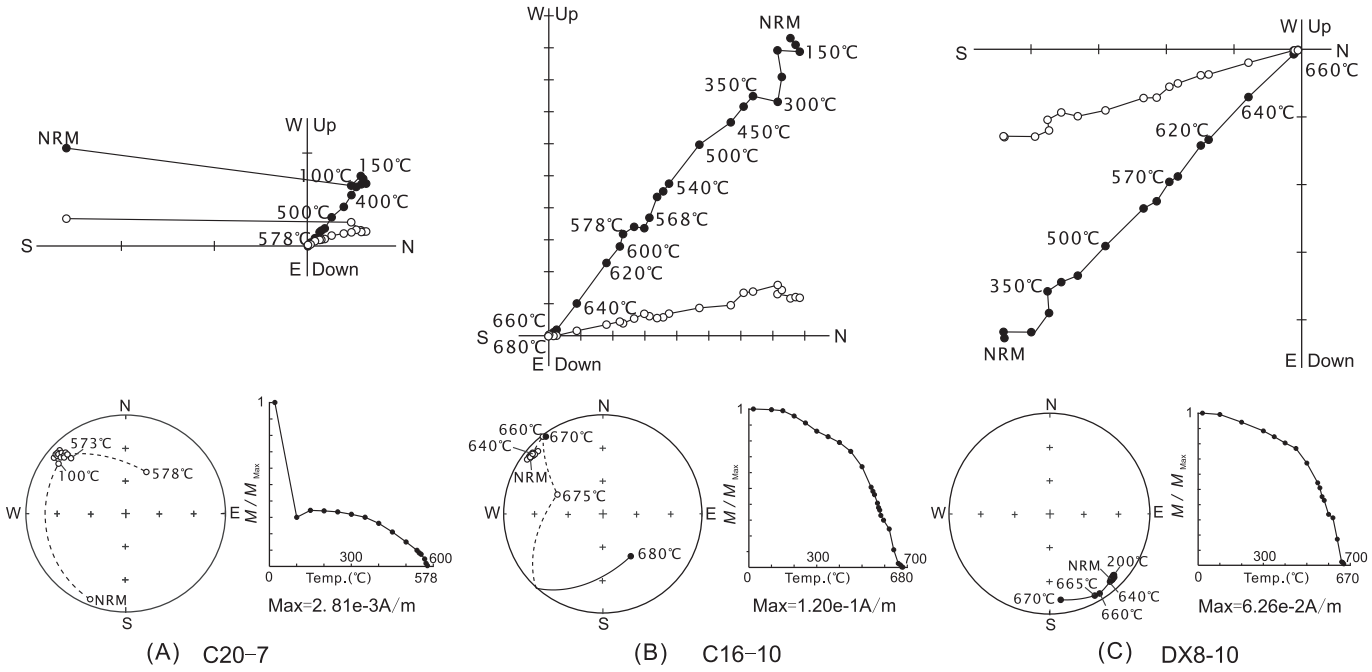


Fig. 3. Orthogonal vector projections of the representative specimens in geographic coordinates. Solid (open) symbols represent the projections onto the horizontal (vertical) plane.

The back-field demagnetization of the SIRM curve shows the largest coercivity at 500 mT (Fig. 4B). These characteristics suggest the presence of high and low coercivity magnetic carriers. Thermal demagnetization of the three-axis IRM shows strong evidence for low-coercivity magnetite unblocked at 580 °C and for high-coercivity hematite unblocked at 680 °C (Fig. 5B). These results reveal that the HTC of lava specimens is carried by magnetite and hematite.

Table 1 lists each site-mean direction of the HTC and the corresponding virtual geomagnetic pole (VGP) calculated. With the exception of site DX12, all sites have similar or antipodal HTC directions after tectonic corrections. It is noted that all HTC directions from site DX12 (Table 1; Fig. 6) and for 4 specimens (DX13) just adjacent to site DX12 do not pass the Jackknife test and are proved to be outliers (Cogné, 2003). They could have been magnetized during a geomagnetic excursion, and therefore have been excluded from the final (site) mean. The site-mean directions for 18 sites in geographic coordinates are Dg = 318.5°, Ig = −2.1°, kg = 22.6, α_{95} = 7.4°, which is significantly different from the present-day field direction (D = 0.4°, I = 46.2°). After bedding corrected, the site-mean direction is Ds = 327.0°, Is = 35.7°, ks = 59.3, α_{95} = 4.5°. The HTC directions pass both McElhinny (1964) and McFadden (1990) fold tests at 99% confidence level (Table 1). Furthermore, the reversal test is positive at the 95% confidence level (McFadden and McElhinny, 1990) (Fig. 6, Table 1). Thus, the HTC is interpreted as a primary magnetization, and its corresponding paleopole should represent Early Cretaceous pole of the study area. Based on the paleomagnetic analysis mentioned earlier, we propose a new Early Cretaceous paleopole for the Lhasa terrane, located at λ = 58.2° N, φ = 341.9° E (A_{95} = 4.6°), indicating a paleolatitude of 19.8° ± 4.6° N for the study area (reference point: 31.3° N, 85.0° E).

Comparing the VGP scatters with those expected by McFadden et al. (1991) is a generally accepted approach to evaluate averaging of secular variation (e.g., Sun et al., 2008; van Hinsbergen et al., 2008). It should be noted that the VGP scatter is not only dependent on latitude, but also strongly relates with the reversal rate (McFadden et al., 1991). The Early Cretaceous Zenong Group volcanic rocks erupted during ca. 110–130 Ma (Zhu et al., 2008). The reversal rate during 110–130 Ma should be slightly faster than that during 80–110 Ma (the Cretaceous Normal Polarity Superchron), but much slower than that during 130–195 Ma (Gradstein et al., 2004). Therefore, the VGP scatter for the Early Cretaceous Zenong Group volcanic rocks (Paleolatitude at 19.8° ± 4.6° N) should be higher than

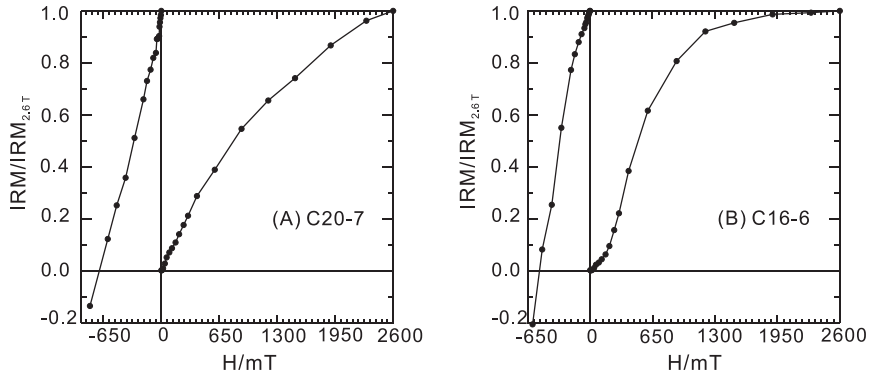


Fig. 4. Isothermal remanent magnetization (IRM) acquisition curves of typical samples, showing the presence of both high- and low-coercivity magnetic minerals.

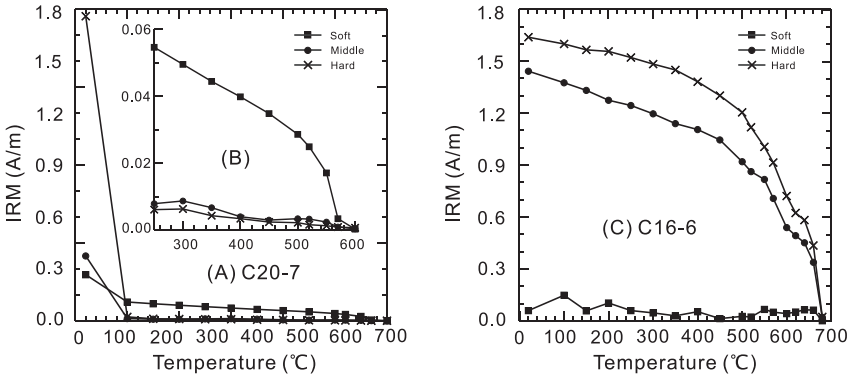


Fig. 5. Thermal demagnetization of a three-component IRM of the representative specimens. The soft (0.12 T), medium (0.4 T), and hard (2.4 T) components are shown during thermal demagnetization.

$S_\lambda = 8.7$ for the 80–110 Ma interval and much lower than $S_\lambda = 18.5$ for the 110–195 Ma interval (McFadden et al., 1991). Based on Fig. 8 of McFadden et al. (1991), the S_λ expected for 110 Ma, 120 Ma, and 130 Ma is estimated at 9.5, 11.6, and 13.6, respectively. Our results show that the Zenong Group volcanic rocks have a VGP scatter of 10.8 that reasonably corresponds to the aforementioned predicted values

Table 1
Site mean values and statistical parameters for the Zenong Group volcanic rocks in Cuoqin area.

Site	n/N	Coor.	Direction			α_{95}/(°)	Paleopole			
			D/(°)	I/(°)	k		Plat/(°N)	Plon/(°E)	dp/(°)	dm/(°)
(1) Site C15-20, DX1-10 (31.37° N, 85.1° E)										
C15	9/9	G	319.2	−2.3	29.8	9.6	39.5	322.9	4.8	9.6
		S	328.3	27.3	29.8	9.6	56.5	332.1	5.7	10.5
C16	6/6	G	321.5	−4.2	36.9	11.2	40.4	319.9	5.6	11.2
		S	333.1	32.0	36.9	11.2	61.9	331.5	7.1	12.6
C17	7/8	G	323.7	−6.3	47.0	8.9	41.2	316.9	4.5	8.9
		S	333.2	29	46.9	8.9	60.9	328.3	5.4	9.8
C18	10/11	G	337.0	0.3	21.0	10.8	51.9	304.4	5.4	10.8
		S	347.7	25.9	21.0	10.8	69.0	300.4	6.3	11.7
C19	7/9	G	315.6	−8.0	24.2	12.5	34.9	323.4	6.3	12.6
		S	323.9	39.9	24.2	12.5	56.9	348.9	9.0	15.0
C20	11/13	G	133.7	−0.2	26.0	9.1	−36.2	148.7	4.6	9.1
		S	145.3	−35.5	26.0	9.1	−56.7	162.9	6.1	10.5
DX1	9/10	G	312.0	−13.6	147.5	4.3	30.3	323.8	2.2	4.4
		S	317.2	37.1	148.1	4.2	50.4	350.1	2.9	4.9
DX2	11/11	G	318.5	−18.7	70.7	5.5	32.9	316.2	3.0	5.7
		S	321.4	29.8	71.0	5.5	51.7	340.5	3.4	6.1
DX3	7/7	G	307.9	−13.4	44.6	9.1	27.3	327.0	4.7	9.2
		S	312.6	38.7	44.5	9.1	46.9	354.3	6.4	10.8
DX4	11/11	G	306.3	−4.3	119.5	4.2	29.0	332.2	2.1	4.2
		S	316.6	47.2	119.3	4.2	52.5	2.0	3.5	5.4
DX5	8/9	G	326.7	−1.4	89.9	5.9	45.0	316.0	3.0	5.9
		S	341.0	36.3	89.7	5.9	69.6	326.4	4.0	6.9
DX6	10/10	G	323.7	−4.3	38.0	7.9	41.9	317.8	4.0	7.9
		S	335.8	36.1	38.0	7.9	65.5	333.4	5.3	9.2
DX7	10/12	G	315.5	−7.2	307.1	2.8	35.1	323.9	1.4	2.8
		S	323.8	39.7	310.0	2.7	56.7	348.8	1.9	3.2
DX8	10/10	G	150.1	2.2	40.5	7.7	−46.9	131.9	3.9	7.7
		S	162.7	−35.9	40.7	7.7	−70.7	142.8	5.2	8.9
DX9	10/10	G	316	−8	145.5	4.0	35.2	323.1	2.0	4.0
		S	324.7	37.6	145.2	4.0	56.9	345.7	2.8	4.7
DX10	12/12	G	314.8	−12.5	115.6	4.1	32.7	322.1	2.1	4.2
		S	320.2	30.4	115.4	4.1	50.9	342.0	2.5	4.6
(2) Site DX11-13 (31.31° N, 84.8° E)										
DX11	9/10	G	305.4	32.6	51.0	7.3	39.0	352.8	4.7	8.3
		S	307.3	30.2	51.1	7.3	40.0	350.0	4.5	8.1
DX12*	9/9	G	94.4	2.2	67.9	8.2	−3.2	171.6	4.1	8.2
		S	94.3	2.5	67.9	8.2	−3.0	171.5	4.1	8.2
DX13	5/10	G	325.5	41.8	145.1	6.4	58.7	350.1	4.8	7.8
		S	329.3	44.7	145.0	6.4	62.7	352.1	5.1	8.1
Mean	162/178	G	318.5	−2.1	22.6	7.4	39.3	323.9	$A_{95}=5.0°$	
	N = 18	S	327.0	35.7	59.3	4.5	58.2	341.9	$A_{95}=4.6°$	

Notes: Lat and Long: latitude and longitude of the locality; n/N: number of samples used to calculate mean and measured; Dg, Ig, Ds, Is: directions in geographic (stratigraphic) coordinates; k: the best estimate of the precision parameter; a_{95}: half angle of the 95% confidence cone on the mean direction. DX12*: proved to be a outlier (Jackknife test); Mean: The site-mean directions of 18 sites excluding the site DX12. ① fold test is positive at the 99% confidence (McElhinny, 1964): Ks/Kg = 2.6 > F(2*(n2 − 1), (n1 − 1)) at 1% point = 2.26; ② fold test is positive at the 99% confidence (McFadden, 1990): critical Xi at 99% = 6.919. Xi1 IS = 7.65, Xi1 TC = 1.03; ③ the classification of reversals tests is C (McFadden and McElhinny, 1990): the normal polarity: N 1 = 16, D1 = 326.2, I1 = 35.6; the reversal polarity N2 = 2, D2 = 154.0, I2 = −36.0; $\gamma_0 = 0.126 < \gamma_{critical} = 0.206$.

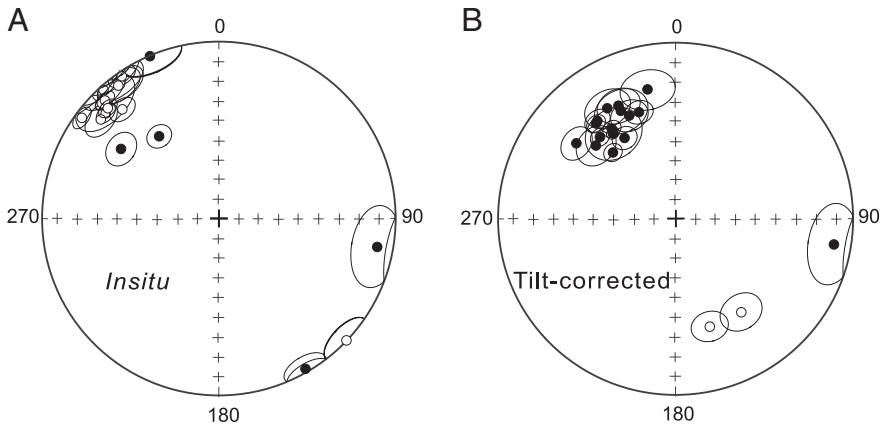


Fig. 6. Equal-area projections of paleomagnetic directions: (A) site mean HTC in geographic coordinates; (B) site mean HTC in stratigraphic coordinates.

based on McFadden et al. (1991). This, combined with that the HTC directions include dual polarities and pass the positive reversal test at the 95% confidence level, suggests that the site-mean direction of the Early Cretaceous Zenong Group volcanic rocks has averaged out secular variation.

Although volcanic rocks and fired archeological materials provide only a series of "snapshots" of paleofield behavior, they can acquire thermal remanence during cooling and in the presence of the magnetic field, and thus are the most accurate paleomagnetic recorder (Tauxe, 1993; Coe et al., 2004). A widely accepted view is that the well-dated lavas which have averaged out secular variation would define a robust paleomagnetic pole. Therefore, our new Early Creataceous result from the Zenong Group volcanic rocks, which have been dated at ca. 110–130 Ma and have averaged out secular variation, is more reliable than those from sedimentary rocks which may be biased by a variety of depositional and post-depositional processes (King, 1955; Tauxe, 2005; Tauxe et al., 2008). Tan et al. (2010) showed that the sedimentary paleomagnetic results without inclination correction could yield spuriously low paleolatitude estimates. Indeed, this is probably the primary reason for such disparate intra-continental shortening estimates calculated from Asian paleomagnetic data (Dupont-Nivet et al., 2010b; Lippert et al., 2011). We have noticed that the Elongation/Inclination (E/I) method developed by Tauxe and Kent (2004) has been applied to correct the magnetic directions and evaluate the possibility of inclination shallowing (Dupont-Nivet et al., 2010a; Tan et al., 2010; Lippert et al., 2011). The E/I technique requires large numbers ($n \geq 100$) of magnetic directions and is based on statistical models of geomagnetic paleosecular variation (Tauxe and Kent, 2004; Tauxe, 2005). Tan et al. (2010) provided a successful case that the E/I corrected inclinations from the Late Cretaceous red beds are consistent with the observed inclinations from the Late Cretaceous lavas.

5. Discussion

5.1. Discrepancy between the Cretaceous poles from the Lhasa terrane

In recent years, the apparent polar wander paths for Asia and India have been reconstructed by Besse and Courtillot (2002) and by Torsvik et al. (2008). Though the paleopoles for Asia and India during the 110–130 Ma got by Besse and Courtillot (2002) are basically consistent with those got by Torsvik et al. (2008), considering that Torsvik et al. (2008) updated paleomagnetic data published between 2002 and 2007 and used more rigorous selection criteria, their complied poles have been used as the reference poles for Asia and India in the present study. Furthermore, van Hinsbergen et al. (2008) reported a well-dated lava-based Early Cretaceous pole from Mongolia, and argued that the 124–94 Ma pole (80.8° N, 158.4° E, $A_{95} = 2.5°$), which has averaged out secular variation, is one of the statistically best-determined paleopoles for Asia. Therefore, this new Early Cretaceous pole from Mongolia has also been combined into the reference pole for Asia during 110–130 Ma.

Several paleomagnetic investigations of Cretaceous rocks from the Lhasa terrane have been carried out in the past three decades (Pozzi et al., 1982; Westphal et al., 1983; Achache et al., 1984; Lin and Watts, 1988; Chen et al., 1993; Sun et al., 2010; Tan et al., 2010). However, significant discrepancies exist between them. We list all available data which were obtained under modern paleomagnetic technological condition (Table 2), and apply the well accepted 7-point data quality criterion system (Van der Voo, 1990) to appraise them. Although this selection criteria are rigorous, most of the results have a Q-value of five or greater. Many poles fail to meet all 7 criteria because they overlapped with the Cenozoic poles from Tibet. In particular, poles QL and NL satisfy three points of the Van der Voo's criterion system (Van der Voo, 1990): pole QL satisfies criteria 1 (well determined rock age), 3 (step-wise demagnetization), and 5 (structural control coherence with the craton discussed), while pole NL satisfies criteria 1 (well determined rock age), 2 (sufficient number of samples), and 3 (step-wise demagnetization). Furthermore, poles NL and QL were defined by a small number of specimens. For these reasons, we ignore them in further discussion.

In order to discuss the lithospheric shortening within the Asian plate and the width of the Neotethys during the Early Cretaceous, Zhongba city (29.77° N, 84.03° E) just located on the India–Asia suture zone has been chosen as the reference location to calculate paleolatitudes. We plot the most reliable poles in Fig. 7. They fall roughly along a small circle centered on the reference location, which indicates relative rotation has occurred between the sampled localities. Furthermore, the paleolatitude discrepancy is also significant (Fig. 8). Tan et al. (2010) recently demonstrated that upper Cretaceous red beds in the Linzhou basin have been biased by inclination shallowing. Notably, the sedimentary units yield the shallowest inclinations, and therefore, the lowest latitude estimates (Fig. 8).

Ages of the Cretaceous poles may cover the time of ca.130–65 Ma (Table 2). The Zenong group was dated to ca. 130–110 Ma using U–Pb geochronology of single zircons (Zhu et al., 2008). However, abundant reversed magnetization directions observed in our sites suggest that the sampled rocks are likely older than the Cretaceous normal polarity superchron C34 (ca. 125–83.5 Ma) (Gradstein et al., 2004). That is, they may be ca. 130–125 Ma. In the Linzhou area of the eastern Lhasa terrane, the Shexing Formation contains the fossils of Albian stage and is covered by the Linzizong Group that yielded a plateau-age of 64.4 ± 0.6 Ma (1σ error) by $^{40}Ar/^{39}Ar$ step heating of a plagioclase (Zhou et al., 2004). The Shexing Formation also yields abundant reversed directions (Tan et al., 2010), which suggest the magnetization

Table 2
Summary of the Cretaceous paleomagnetic poles from the Lhasa terrane (reference point: 29.77° N, 84.03° E).

Mnemonic	Rock-units	Age (Ma)	Plat (°N)	Plon (°E)	A_{95}[dp/dm] (°)	Palat (°N)	N	Criterion (Q)	Reference
QL	Qelico lava	85–95, ^{40}Ar–^{39}Ar	74.0	318.0	11.1/19.1	19.7 ± 14.6	20	1□3□5□□ (3)	Lin and Watts (1988)
NL	Nagqu lava	95–100, ^{40}Ar–^{39}Ar	78.0	282.0	4.0/6.9	18.3 ± 5.3	33	123□□□□ (3)	Lin and Watts (1988)
SR*	Shexing Fm, red beds, Linzhou	Albian–Late Cretaceous	79.6	330.5	1.9/3.1	25.2 ± 2.4	196	123F5D□ (6)	Tan et al. (2010)
SL	Shexing Fm, lava, Linzhou	Albian–Late Cretaceous	69.1	191.7	3.3/5.4	21.7 ± 4.2	136	123F5D□ (6)	Tan et al. (2010)
TP	Takena Fm,red beds, Linzhou	Albian–Aptian	68.0	340.0	6.7/11.6	22.4 ± 8.8	68	123 F5□7 (6)	Pozzi et al. (1982)
TW	Takena Fm, red beds,Linzhou	Albian–Aptian	64.0	348.0	5.6/9.5	24.0 ± 7.3	57	123F5D7 (7)	Westphal et al. (1983)
TNA	Takena Fm, red beds, North region	Albian–Aptian	63.5	325.4	6.5	15.0 ± 6.5	57	123 F5□7 (6)	Achache et al. (1984)
TSA	Takena Fm, red beds, Linzhou	Albian–Aptian	71.2	288.4	7.9	12.4 ± 7.9	61	123 F5□□ (5)	Achache et al. (1984)
TL	Takena Fm, red beds, Linzhou	Albian–Aptian	68.0	279.0	3.5/6.9	8.4 ± 4.9	51	123 F5□□ (5)	Lin and Watts (1988)
KWT	Combine pole, Cretaceous sandstone and limestone, western Tibet	Albian–Aptian	66.2	245.0	5.1	7.1 ± 5.1	104	123F5D7 (7)	Chen et al. (1993)
WR	Woronggou Fm, volcanic rocks, Deqing	114 ± 1.1, U–Pb SHRIMP	66.4	220.3	6.9	11.8 ± 6.9	88	123□5□7 (5)	Sun et al. (2008)
ZN	Zenong Gp, lava, Cuoqin	110–130, U–Pb LA ICP MS	58.2	341.9	4.6	19.0 ± 4.6	162	123F5R7 (7)	This study

Notes: Mnemonic: pole mane plotted in Fig. 7 and used in the text; age of the rock-unit: number being of isotopic age followed by the method, geological stages based on regional stratigraphic correlations; Plat and Plon: latitude and longitude of the pole; A_{95}: half angle of the 95% confidence cone of the pole; dp/dm: semi-axes of elliptical error of the pole at a probability of 95%; Palat: Palaeolatitude; N: sample number; Criterion (Q): data quality criteria and Q values (number of criteria met) after Van der Voo (1990) [1, well determined rock age; 2, sufficient number of samples, n>24, k (or K) ≥ 10, $\alpha_{95} \leq 16.0^{\circ}$; 3, step-wise demagnetization; 4, field tests; 5, structural control and tectonic coherence with the craton discussed; 6, presence of reversal; 7, no resemblance to paleopole of younger age (by more than a period)], numbers mean satisfying these criteria, of which for criterion 4, F means positive fold test, for criterion 6, R means positive reversals test; D means dual-polarity; "□": failed to meet this criterion. *Based on E/I corrected inclination data.

formed after the C34. It is likely Late Cretaceous in age. The Takena Formation conformably underlies the Shexing Formation and several investigations revealed only a few reversed directions, so the age of the KWT pole in Table 2 is also consistent with the Aptian–Albian stages that are within the C34. If this analysis can be accepted, the youngest pole SL and the oldest pole ZN stand as two ends of the distribution of all the Cretaceous poles from the Lhasa terrane (Fig. 7). If we further combine the poles SR and SL, because they are from the intercalated lava and red beds of the Shexing Formation, their average pole falls close to the stable Asia reference poles of ca. 80–60 Ma. However, the pole ZN is significantly from the coeval Asia reference pole of ca. 110–130 Ma. We hence suggest a ca. 50° anticlockwise rotation between ca. 120 Ma and 65 Ma happened for the Lhasa terrane. Alternately, this anticlockwise rotation could be caused by a local rotation. However, further paleomagnetic studies, as well as geological evidence, are necessary to confirm the rotation discrepancy in the future.

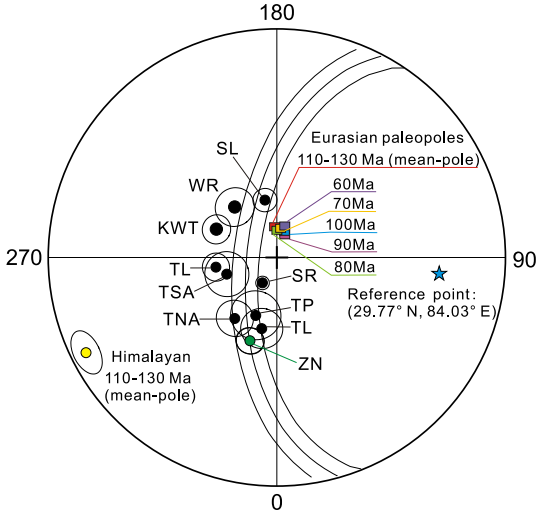


Fig. 7. Equal-area projections of the Cretaceous paleomagnetic poles of the Lhasa terrane with reference locations, respectively. See Table 2 for values and references. Small arcs: small-circles passing through reference locations; Blue star: reference location (29.77° N, 84.03° E); red rectangle: the mean pole for stable Asia combined the poles got by Torsvik et al. (2008) and van Hinsbergen et al. (2008) during 110–130 Ma; other colourful rectangle: referencce poles for stable Asia got by Torsvik et al. (2008) during 60–100 Ma; yellow dot: the mean pole for India during 110–130 Ma (Torsvik et al., 2008).

5.2. Crustal shortening within Asia

According to geological evidence (Dewey et al., 1988; Chen et al., 1993; Matte et al., 1996; Metcalfe, 1996, 2006; Halim et al., 1998; Wu et al., 2002; Kapp et al., 2005, 2007; Guynn et al., 2006; DeCelles et al., 2007), the Lhasa and Qiangtang terranes are separated by the Late Jurassic–Early Cretaceous Bangong–Nujiang suture, and the Lhasa terrane became a part of stable Asia before the Cretaceous. Furthermore, the combined robust paleomagnetic data from the 60 to 40 Ma of the Lhasa terrane indicate that the southern margin of Asia (reference point at 31.3° N, 85.0° E) was located at 18.2° ± 8.5° N (Achache et al., 1984; Chen et al., 2010; Dupont-Nivet et al., 2010a; Liebke et al., 2010; Sun et al., 2010; Tan et al., 2010) which is similar to the Cretaceous paleolatitude (19.8° ± 4.6° N) of the same reference point, suggesting that the Lhasa terrane experienced no significant

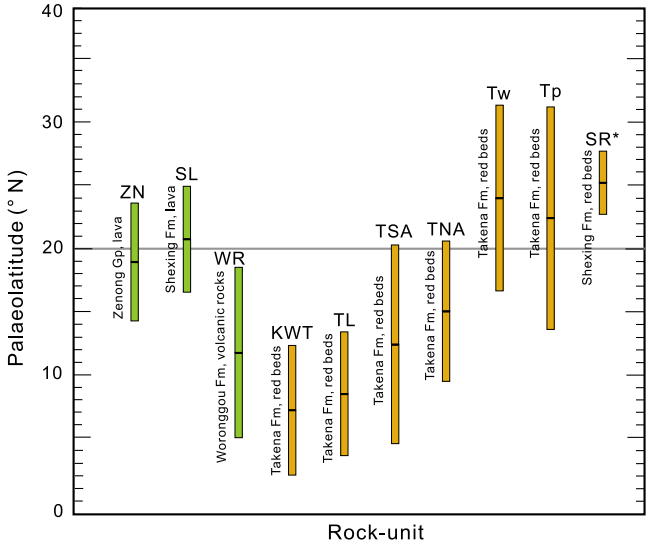


Fig. 8. Plot of Cretaceous paleolatitudes of the Lhasa terrane (reference point: 29.77° N, 84.03° E). Gp: group; Fm: formation (details see Table 2). Uncertainties of paleolatitudes are denoted by vertical bars.

latitudinal motion between Cretaceous and Paleogene time. Comparing our observed Early Cretaceous pole with the mean paleopole for Asia between 110 and 130 Ma ($\lambda = 79.1°$ N, $\varphi = 186.9°$ E, $A_{95} = 4.7°$) (Torsvik et al., 2008; van Hinsbergen et al., 2008) indicates a paleolatitude difference of $7.8° \pm 4.7°$, implying that ca. 870 ± 520 km north–south crustal shortening has taken place between the Lhasa terrane and stable Asia since the Early Cretaceous (Fig. 7).

The paleomagnetic data summarized earlier is well consistent with the post-Early Cretaceous geologic record of Asia that suggests a total of ca. 870 km of crustal shortening has been absorbed by fold and thrust belts and strike–slip faults between the Lhasa terrane and stable Asia. Abundant geological and geophysical evidence indicates that tectonic activity in Asia is the consequence of continental collision between India and Asia (Hendrix et al., 1994; Molnar and Tapponnier, 1975; Patriat and Achache, 1984; Zhang et al., 1996; Meyer et al., 1998; Xu et al., 1999a,b; Ge and Liu, 1999; Yin and Harrison, 2000; Jolivet et al., 2001, 2010; Wu et al., 2002; Darby et al., 2005; Wang et al., 2008; Dayem et al., 2009; Duvall et al., 2011). This tectonic activity includes intracontinental shortening between the southern margin of the Lhasa terrane (Fig. 1) and the Qiangtang terrane; in the Kunlun Shan, Qilian Shan, Tian Shan, Altai, and other mountain ranges; intracontinental subduction beneath Qaidam and Tibet; and strike–slip faulting along the Altyn Tagh, Karakorum (Coward et al., 1988; Meyer et al., 1998; Ge and Liu, 1999; Xu et al., 1999b; Yin and Harrison, 2000; Wu et al., 2002; Darby et al., 2005; Yue et al., 2005; Yin et al., 2007; Wang et al., 2008; Lippert et al., 2011). Yin and Harrison (2000) reviewed the geologic history of the Himalayan–Tibetan orogene and suggested that at least 1400 km of north–south (N–S) has been absorbed by the orogenic belts since the onset of the India–Asian collision at about 70 Ma. About 950 km N–S shortening has been absorbed by thrust belts (Fig. 1), which include the Shiquanhe–Gaize–Amdo thrust system (ca. 250 km), the Fenghuo Shan–Nangqian fold and thrust belt (ca. 70 km), the Qimen Tagh–North Kunlun thrust system (ca. 270 km), and the Nan Shan thrust belt (ca. 360 km), between the sampling area and Hexi corridor (North China Block).

5.3. The width of the Neotethys during the Early Cretaceous

Using the average pole ($\lambda = 7.5°$ N, $\varphi = 296.4°$ E, $A_{95} = 6.2°$) of the India apparent polar wander path during 110–130 Ma (Torsvik et al., 2008), we calculate a paleolatitude of $41.4° \pm 6.2°$ S for a reference point located at 29.77° N, 84.03° E stable Indian coordinates (Fig. 7). Comparing the paleolatitudes between cratonic India and the Lhasa terrane indicates that the Neotethys, which separated the India and Lhasa terrane, was 6720 ± 690 km wide in the Early Cretaceous. Li et al. (2002) suggested that the Neotethys gradually began to open after the Middle Triassic and the oceanic basin had reached to its greatest width by the Early Cretaceous. Subsequently, the Neotethys became smaller as it was subducted beneath the Lhasa terrane in Late Cretaceous to early Paleogene time. Thus, our estimate for the width of the Early Cretaceous Netotethys is consistent with geologic and geophysical opinions.

6. Conclusions

We have collected paleomagnetic data from lower Cretaceous volcanic rocks from the Lhasa terrane. Site-mean directions pass a fold test at high confidence. Our new paleomagnetic results when combined with the previous paleomagnetic data from the Lhasa terrane, show that the Lhasa terrane maintained a stable latitude from the Early Cretaceous to the early Paleogene. The difference of the paleolatitudes between our observed directions and the expected direction for stable Asia implies that ca. 870 km north–south crustal shortening had accumulated between the Lhasa terrane and stable Asia since Early Cretaceous. Furthermore, comparing our paleomagnetic data with the India apparent polar wander path suggests the Neotethys was 6720 ± 690 km wide during the Early Cretaceous.

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